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EDUCATION

National University of Singapore

M.Tech. in Software Engineering

Aug 2026 – Nov 2027

Singapore

University of Electronic Science and Technology of China

B.Eng. in Software Engineering

Sep 2020 – Jun 2024

Chengdu, China

EXPERIENCE

Virtualization R&D Engineer

SmartX

Jul 2024 – Sep 2025

Chengdu, China

- **Migration product:** led the design of V2V OS 1.6.0 to 1.7.2, which moves virtual machines from VMware to SMTX OS and between SMTX clusters. Built across the end-to-end path: VMware CBT snapshots, a Python control plane, and a C data-transfer core exposed as a shared library. The product now serves 10,000+ production VMs, including core financial workloads.
- **Transfer performance:** replaced the serial read-then-write path with asynchronous chunked reads and writes across a bounded queue, using 8 reader and 8 writer workers. Handled partial I/O, cancellation and first-error propagation across the pipeline. Increased sustained throughput from 70 MB/s to 290 MB/s.
- **VMware compatibility:** isolated each migration in its own process, with one VDDK version loaded per process. This removed shared-process SDK symbol and global-state conflicts, contained crashes to one task, and expanded support across vSphere releases.
- **Destination-platform support:** prepared migrated VMs to boot on new virtual hardware through offline driver injection. Replaced the NFS disk-attachment path after hung mounts blocked guest preparation, using iSCSI block devices instead; then adapted the pipeline to SMTX ELF disaggregated compute-storage clusters.
- **In-guest agent:** rebuilt VMTools, replacing shell scripts over QGA guest-exec with native QEMU Guest Agent commands and resource probes. Cut response time from 3 s to under 50 ms and yearly escalations from 40 to under 5.
- **Consistent snapshots:** made snapshots filesystem-consistent rather than only crash-consistent. The control plane coordinates VMTools and fsfreeze to quiesce the guest filesystem before the storage snapshot, then thaws it after the snapshot boundary.
- **Cross-cluster live migration:** built a block-level data check triggered by a libvirt hook after switchover. It snapshots source and destination disks through the storage API, reads and hashes them block by block, and keeps compare order in sequence queues on both sides so verification runs concurrently without blocking migration.
- **Kernel and QGA debugging:** used strace to find a UOS kernel creating pids without reaping them, hanging customer VMs; the fix became a financial-sector advisory and earned a written commendation. Also traced truncated host-to-guest JSON to old kernels not re-polling the QGA serial fd.

Virtualization R&D Intern

SmartX

Oct 2022 – Jul 2024

Chengdu, China

- **Migration appliance:** built V2V OS 1.2.0 to 1.6.0 as a self-contained virtual appliance on openEuler for government procurement. Rebuilt and packaged its kernel with the filesystem and block-device features required to inspect and modify newer guest images.
- **Low-downtime migration:** extended the workflow from one full copy plus a long outage to an initial copy followed by repeated incremental passes while the VM keeps running. VMware Changed Block Tracking (CBT) identifies blocks dirtied after each snapshot, leaving only a small final delta for the stopped-VM switchover.
- **Packaging and release:** wrote the rpm.spec for V2V and its custom kernel. Built CI/CD that assembles both into a bootable ISO and a Packer pipeline that produces the vSphere OVF appliance.
- **Offline driver injection:** made migrated guests boot on virtio hardware by rebuilding Linux initramfs images and replaying Windows registry changes captured with procmon around a virtio-win installation. Supported both MBR and UEFI, with disk snapshots and an IDE fallback for recovery.
- **Guest coverage:** carried driver injection and VMTools across nine Linux distributions, including openEuler, UOS and Kylin, plus Windows Server 2012–2025. Maintained per-OS handling for boot layout, package tools, service startup and guest-agent installation.

Systems Integration Development Intern

ByteDance (Feishu)

Jun 2022 – Sep 2022

Chengdu, China

- **Workflow integration:** built Go and Gin services connecting recruiting workflows to external OA systems, with MongoDB holding sync and retry state.
- **API hardening:** added authentication, rate limiting and a regression suite across the integration endpoints.

PROJECT

Maintainer

Sep 2026 – Present

FastMM, market-making engine in C++20

ziy.bio/FastMM

- **C++ hot path:** built one template-policy/CRT engine for backtest, replay and live trading with no virtual calls, heap allocation, exceptions or network waits in the trading loop. Used int64 fixed point, a pre-faulted arena, cache-line-aligned SPSC rings, pinned cores and busy spinning; simulated book-update-to-order latency is 111 ns p50 and 167 ns p99.
- **Kernel-bypass market data:** implemented AF_XDP with UMEM, FILL/RX rings and an XDP/BPF multicast filter, supporting zero-copy, native-copy and generic modes. Added DPDK poll-mode burst RX with preallocated mbufs and a kernel exception tap.
- **Exchange protocols:** built Nasdaq ITCH ingestion over redundant MoldUDP64 A/B feeds, with reordering, gap re-requests, GLIMPSE recovery and L3/L2 books. Added OUCH, FIX and CME MDP3 codecs plus live Binance, Bybit, OKX and Deribit connectors.
- **Latency engineering:** measured receive-to-send stages with rdtscp-based T0–T5 histograms. Implemented and benchmarked epoll and multishot io_uring reactors, keeping epoll as the default when TCP read/write costs dominated.
- **Correctness and safety:** journaled every event so replay emits byte-identical orders; reconciled fills, orders and positions after disconnects or restarts. Added pre-trade risk checks, feed-lag gates, kill switches, L2 queue-position models and walk-forward validation.

RESEARCH

Researcher

Sep 2025 – Aug 2026

HKUST (Guangzhou), LLM systems: agent memory, decoding, interpretability

Guangzhou, China

Under review

- Z. Liu, Y. Shi. A Simpson's Paradox in In-Context Learning Curves. ICLR 2027 submission. The average loss falls but rises within every token-occurrence group; domain explains 99% of the standard score's variance, versus 0.2% for model size.
- Y. Shi, Z. Liu, K. Luo, N. Tang. A Living In-the-Wild Benchmark for Measuring the Static-Benchmark Gap in AI-Generated Video Detection. ICLR 2027 submission. On VidTide's 21,504 clips, five detector families fall from published 93–96% AUROC to 52–71%.
- Y. Shi, K. Luo, Z. Liu, N. Tang. Neuro-Symbolic Forensic Reasoning for Open-Generator AI-Generated Video Detection. ICLR 2027 submission. Without training detector weights, FOVEA reaches 97.39% accuracy and 0.99 AUROC on ten unseen generators.

In preparation

- **Committed SAE-feature traces:** made a hosted API prove it served the model it advertised. Fingerprinted every answer with a sparse-autoencoder feature trace, committed to a Merkle root before any check is requested. Cheating then means failing the check or paying for the real model anyway. Caught all 17 substitutes across three backbones, under 2.1% overhead.
- **Cooperative memory paging:** kept a long chat usable after it outgrows the context window. Checked first whether the model signals a missing fact through token-level uncertainty. It does not, and answers more confidently without it. Left an ~8-token keyword bookmark in place of each dropped section, plus a recall() tool for the full text. Beat five methods on LoCoMo, four LLM judges agreeing at $p = 0.017$.
- **Copy-as-decode:** cut the cost of re-emitting unchanged text when an LLM edits a file. Had it name the input lines to copy, spliced into the KV cache by one parallel prefill. Speculative decoding, except the draft is the input document and a grammar guarantees acceptance. Copied text is 74 to 98% of an edit, bounding the speedup at 13×.

TECHNICAL SKILLS

Languages: Python, C++, C, Rust, Go, Shell

Networking: TCP/UDP multicast, epoll, io_uring, AF_XDP, DPDK, tcpdump

Linux/UNIX: processes and threads, virtual memory, NUMA, CPU affinity, cgroups, strace/perf/gdb

Virtualisation & migration: QEMU/KVM, libvirt, Virtio, VMware CBT/VDDK, QGA, live migration

LLM systems: memory tiering, retrieval and reranking, ReAct, MCP, KV cache

Build & operations: CMake, GitHub Actions, Jenkins, Docker, RPM/DEB, Packer

Databases: MySQL, Redis, MongoDB, SQLite

HONOURS

National Third Prize, China Collegiate Computer Design Competition (2022) | UESTC Scholarship (2020–21)